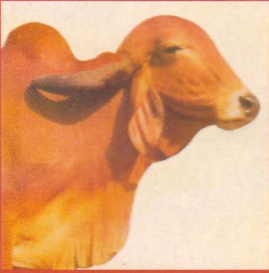


ANIMAL GENETIC RESOURCES OF INDIA



CATTLE

Dr. Vikas Vohra

Dr. Arjava Sharma



भाकृअनुप-राष्ट्रीय पशु आनुवंशिक संसाधन ब्यूरो
ICAR-National Bureau of Animal Genetic Resources



Baldi Bypass, P.O. Box-129, Karnal-132001(Haryana) India

BELAHI

ACCESSION NUMBER

INDIA_CATTLE_0532_
BELAHI_03038

ORIGIN / NATIVE TRACT

State

Haryana and UT of
Chandigarh

District(s) / Region

Panchkula, Ambala,
Yamuna Nagar

POPULATION

~20,000

BREED STATUS

Not at risk

ABOUT THE BREED

Belahi is a dual utility (milch and draught) breed of cattle having medium stature with symmetrical body, tight skin, straight and broad forehead with prominent poll and slender face. Hump is more prominent in males. The cow have potential of moderate milk production and males are used for ploughing the agricultural fields in Shivalik foot hills. The breed management is extensive and is semi-migratory in nature. The breed has higher heat & physical stress tolerance capability. Temperament of the breed is moderate. Semen of Belahi breed is not available for AI, as livestock keepers prefers natural service and always maintains a breeding bull in a herd.

PHYSICAL DESCRIPTION

Coat Colour: Belahi cattle have a reddish coat color having white coloured ventral parts of body. Face and dewlap are white in colour. Muzzle is black.

Horn Shape & Size: The horns are medium in size and sickle shaped.

Distinguishing Characteristic: Unique body color mix of red and white, face and dewlap are white in colored. Straight forehead with prominent poll.

PERFORMANCE

Cows are good milkers and produce an average of 1000 kg of milk in a lactation with a lactation length of 230 days. Age at first calving ranges from 900 to 1400 days and calving interval from 365-400 days with an average of 400 days. The average dry period is reported around 195 days. The service period ranges from 60 days to 140 days. The bulls and bullocks are used for ploughing and can continuously work for 3-3.5 hours without showing the signs of fatigue.

BELAHI



MILCH & DRAUGHT BREED

ACCESSION NUMBER

INDIA_CATTLE_0400_
GIR_03007

ORIGIN / NATIVE TRACT

State
Gujarat

District(s) / Region
Amreli,
Bhavnagar,
Junagadh
Kathiawar region

POPULATION

~ 13,80,208

BREED STATUS

Not at Risk

ABOUT THE BREED

Gir is a milch breed of cattle. Cow has excellent potential for milk production and therefore, the breed can be used for upgrading of non-descript cattle of India. Animals have large sized body with loose body conformation and are comparatively more resistant to tick borne infections. Temperament is docile. Gir has been imported by Brazil, USA and other countries for exploiting its milk production and capability of tolerating high physical environmental stress. Semen of Gir breed is available at AI station Rajkot (Gujarat) and Sabarmati Ashram Gaushalla, Bidaj, Junagadh Agricultural University and also with some cooperative (AMUL) & private dairies of Gujarat.

PHYSICAL DESCRIPTION

Coat Colour: Most of the Gir animals have pure red coat colour. Some animals are speckled red.

Horn Shape & Size: Long horns are peculiarly curved. Starting at the base of the crown they take a downward and backward curve and again incline a little upward and forward taking a spiral inward sweep, finally ending in a fine taper.

Distinguishing Characteristic: Bulging and prominent head, convex forehead. Long and pendulous ears folded like a leaf. Prominent hip bones.

PERFORMANCE

Cows are excellent milkers and have capacious udder. Cow produces about 1100 to 2000 kg of milk in a lactation period of about 300 days. The average lactation yield is around 1560 kg. At organized herds the cows can easily produce around 2000 kg in a lactation. Average age at first calving is 1390 days. The average calving interval is around 520 days. The service period ranges from 116 days to 317 days.



MILCH BREED

HARIANA

ACCESSION NUMBER

INDIA_CATTLE_0520_
HARIANA_03009

ORIGIN / NATIVE TRACT

State

Haryana

District(s) / Region

Hissar, Rhotak,
Jind, Gurgaon
(Widely spread across
Indo-gangetic plains)

POPULATION

~ 16,39,181

BREED STATUS

Not at Risk

ABOUT THE BREED

Hariana is a famous and front liner among prominent dual-purpose (draught and milch) breed of India. It is better known more for draught potential rather than milk production. Animals are medium sized and have well proportionate body. Face is long and narrow with straight forehead. Well marked bony prominence is seen at the poll. Dewlap is small in size. Breed has predominantly been used in upgrading the nondescript cattle particularly to improve the draught ability in almost entire region of Indo-Gangetic plains in Uttar Pradesh along with Rajasthan. Semen of this breed is available at ABC salon and FSBS Babugarh (Uttar Pradesh), Bassi (Rajasthan), and Sabarmati Ashram Gaushala, Bidaj (Gujarat)

PHYSICAL DESCRIPTION

Coat Colour: Animals are white or light grey. In bulls colour in between fore and hind quarters is relatively dark or dark grey.

Horn Shape & Size: Horns are short and small.

Distinguishing Characteristic: White colour, long and narrow face, flat forehead, well-marked bony prominence at the center of poll and small horns.

PERFORMANCE

Udder is well developed. Cows are fairly good producers of milk and produce around 1000 kg which ranges from 700-1,700 kg of milk in a lactation of 250 days. The average first lactation yield is around 800 kg. The average lactation length varies from 230-330 days, with an average of 276 days. Age at first calving is around 1566 days. The average dry period reported is around 250 days. The average calving interval is 527 days. The mean birth weight of calf is around 22 kg and weight at first calving is reported at around 325 kg. Males have excellent draught ability and are extensively used for transport and other agricultural operations.



MILCH & DRAUGHT BREED

RED SINDHI

ACCESSION NUMBER

INDIA_CATTLE_0000_
REDSINDHI_03025

ORIGIN / NATIVE TRACT

Organized farms of Odisha,
Tamil Nadu, Bihar,
Uttarakhand,
Assam and Kerala

POPULATION

~ 59,642

BREED STATUS

Not at Risk

ABOUT THE BREED

Red Sindhi is an important milch breed of cattle. Animals are medium in size and have compact and proportionate body conformation. Hump size is medium to large. The animals of the breed are heavy and considered to have better heat tolerance capability. The cow have potential of good milk production and therefore, the breed has been imported by more than 20 countries including America, Australia, Phillipines, Brazil and Sri Lanka. Temperament of the breed is docile. Semen of the breed is available at Sabarmati Ashram Gaushalla, Bidaj, Cuttack Odisha, DLF Hosur (T.N) and ABC, Salon (U.P) and Rishikesh (Uttarakhand).

PHYSICAL DESCRIPTION

Coat Colour: This breed has distinctly red colour. Red shades may vary in colour. In bulls, colour is dark on the shoulders and thighs.

Horn Shape & Size: Horns are thick at the base and emerge laterally and curve upward.

Distinguishing Characteristic: Large and compact body with straight and short face. Prominent head, loose skin, red body colour, large hump and thick horns at base.

PERFORMANCE

Cows have capacious udder and on an average produce about 1800 kg of milk in a lactation period of about 300 days. The average first lactation yield is around 1500 kg, which varies from 1100 to 2600 kg. Age at first calving is around 1400 days. The average dry period is around 180 days. The service period ranges from 100 to 160 days. The average calving interval is around 385 days.

RED SINDHI



MILCH BREED

SAHIWAL

ACCESSION NUMBER

INDIA_CATTLE_1617_
SAHIWAL_03026

ORIGIN / NATIVE TRACT

State
Punjab

District(s) / Region
Amritsar,
Ferozpur

State
Rajasthan

District(s) / Region
Ganganagar

POPULATION

~ 10,92,459

BREED STATUS

Not at Risk

ABOUT THE BREED

Sahiwal is one of the best dairy breed of cattle. Animals are medium in size having symmetrical and deep body. Sahiwal animals have been imported by many countries to utilize its milk production potential. It is well adapted to high temperature and humidity, and is comparatively more resistant than temperate breeds to ticks and diseases. The germplasm/seed is available with ICAR-National Dairy Research Institute, Karnal, Punjab State Semen Collection Centre, Nabha and also with some cooperative & private dairy in Punjab and Rajasthan.

PHYSICAL DESCRIPTION

Coat Colour: Coat colour is usually reddish dun but pale red or brown occasionally seen in few animals.

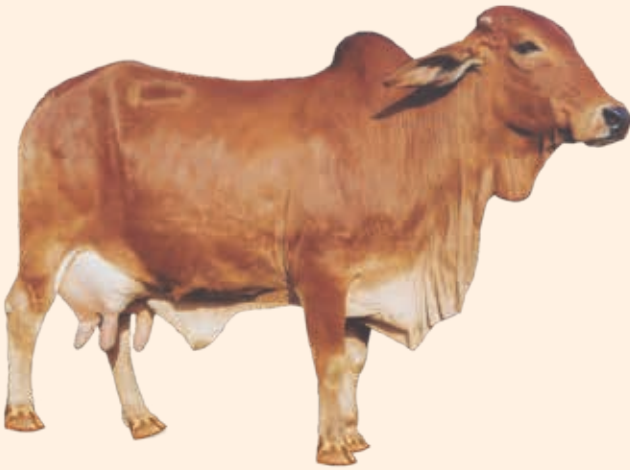
Horn Shape & Size: Horns are short and stumpy, pale red in colour

Distinguishing Characteristic: Large and loose skin, prominent forehead, large Dewlap. Long and pendulous teats in females. Pendulous Navel sheath. Large size of hump in males.

PERFORMANCE

Cows have large and capacious udder and produce around 2,300 kg of milk in a lactation of 300 days. The production varies from 1600 to 2700 kg in one lactation. The average milk production is 2000 kg in a lactation. The average age of first calving is around 1267 days, ranging from 880 to 1580 days. The average service period is around 168 days, ranging from 68 days to 270 days. Average calving interval ranges from 440-480 days with an average of 475 days. Dry period are reported to be around 156 days. The average birth weight reported is around 22 kg in calves. Weight at calving in adult cows ranged from 287 kg to 410 kg.

SAHIWAL



MILCH BREED

THARPARKAR

ACCESSION NUMBER

INDIA_CATTLE_1700_
THARPARKAR_03028

ORIGIN / NATIVE TRACT

State

Rajasthan

District(s) / Region

Barmer,
Jaisalmer,
Jodhpur

POPULATION

~ 1, 97,291

BREED STATUS

Not at Risk

ABOUT THE BREED

Tharparkar is a milch breed of cattle. This breed is native to the desert areas of India. Animals are medium in stature and has compact body conformation. The breed is well adapted to hot and dry agro-climatic conditions, and is resistant to tropical diseases. Semen of this breed is available at semen station Bassi (Rajasthan) and at ABC, Salon (Uttar Pradesh), Durg (Chhattisgarh), CFST&TI Hessarghata (Karnataka), and Central semen station Bhopal (Madhya Pradesh), BAIF Urulikanchan (Maharashtra).

PHYSICAL DESCRIPTION

Coat Colour: Animals are white or light grey. Face and extremities are of a darker shade than the body. In bulls neck, hump, and fore and hind quarters are in dark shade.

Horn Shape & Size: Horns are thicker and short to medium in size. Horns are set well apart curving gradually upward and outward in the same line as that of the poll, with blunt points inclined inward, moderately thick at the base.

Distinguishing Characteristic: White coat colour and large skin of dewlap, broad forehead. Medium to large sized hump.

PERFORMANCE

Cows have good milk potential. Udder is large and capacious. Cows produce around 1,870 kg of milk in a lactation of about 285 days. The lactation milk yield ranges from 1400 to 2,100 kg in a lactation. Average milk yield in first lactation is around 1500 kg in a lactation. The dry period is recorded ranging from 115-191 days with an average of 137 days. The age at first calving is around 1230 days. Average calving interval is 450 days.

THARPARKAR



MILCH BREED

MURRAH

ACCESSION NUMBER

INDIA_BUFFALO_0500_
MURRAH_01001

ORIGIN / NATIVE TRACT

State

Haryana

District(s) / Region

Rohtak, Jind, Hisar, Jhajhar,
Fatehabad, Gurgaon

Union territory

Delhi

POPULATION

1, 16, 86,198

BREED STATUS

Not at risk

ABOUT THE BREED

Murrah is large sized breed of buffalo. Tail is long. Wedge shaped and deep body conformation. Murrah is one of the best dairy breeds of buffaloes in the world and has been taken to majority of the countries where buffalo is now raised for milk and meat either as purebreds or for grading up the local buffaloes. In India, large number of grades of Murrah buffaloes is available in different states especially where this breed had been used as an improver breed for increasing milk production in nondescript buffaloes through upgrading. This breed can be seen throughout the length and breadth of India.

PHYSICAL DESCRIPTION

Coat Colour and Skin: Jet Black / Pure black. Skin is tight, soft and thin.

Horn Shape & Size: Tightly curved in a spiral form. Short in size of horn curl fashion which gives it the name 'Murrah'.

Distinguishing Characteristic: Jet black colour, tightly curled horns/ single coil of horns, Thin Skin, heavy body size.

PERFORMANCE

Females are excellent yielders. Udder is capacious and well-developed. Total lactation milk yield in Murrah buffaloes is around 2,000 kg of milk in about 320 days, but there is a large variation both in size and milk production. Some elite buffaloes may produce as high as 4000 kg milk in a lactation of around 350 days. The maximum daily milk yield reported in Murrah buffaloes is around 26 kg. The average age at first calving is 1550 days. Average inter-calving period is around 450 days. The average fat percentage in milk is around 7 percent. Males and bullocks are sturdy not easily fatigued and are extensively used for draught, transport and other agriculture operations.

MURRAH

